



Chapter #7: Building Blocks of Integrated Circuit Amplifiers

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Introduction

■ IN THIS CHAPTER WE WILL LEARN

- The **basic integrated-circuit (IC) design philosophy** and how it differs from that for discrete-circuit design.
- The **basic gain cells of IC amplifiers**, namely, the CS and CE amplifiers with current-source loads.
- How to **increase the gain** realized in the basic gain cells by employing the principle of cascoding.



Introduction

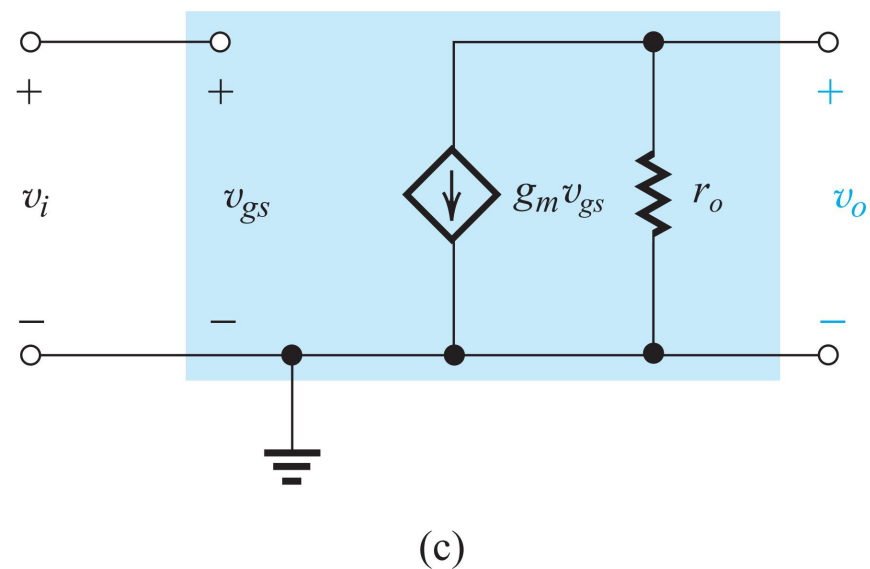
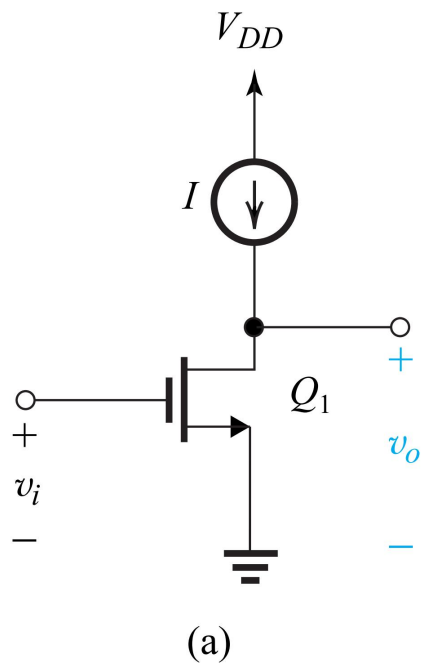
- IN THIS CHAPTER WE WILL LEARN
 - How **current sources are used to bias IC amplifiers** and how the reference current generated in one location is replicated at various other locations on the IC chip by using current mirrors.
 - Some **ingenious analog circuit design techniques** that result in current mirrors with vastly improved characteristics.
 - How to **pair transistors** to realize amplifiers with characteristics superior to those obtained from a single-transistor stage.

7.1. Integrated Circuit Design Philosophy

- **Integrated-circuit fabrication technology** imposes constraints on – and provides opportunities to – the circuit designer.
 - large capacitors are not available
 - very small capacitors are easy to fabricate
- One objective is to realize as **many functions as possible** using MOS transistors only.
- **Reduction of device size** is of great concern.
- In this chapter, focus is placed on CMOS circuit fabrication.

7.2. The Basic Gain Cell

- Two types of basic gain cells exist:
 - Common-source (CS)
 - Common-emitter (CE)
- Both are loaded with **constant-current source**.
 - This is done because of difficulties associated with fabrication of exact resistances.
 - It also facilitates increased gain.
- These circuits are referred to as **current-source loaded / active loaded**.



$$R_{in} = \infty \quad (7.1)$$

$$A_{vo} = -g_m r_o \quad (7.2)$$

$$R_o = r_o \quad (7.3)$$

Figure 7.1 The basic gain cells of IC amplifiers: **(a)** current-source- or active-loaded common-source amplifier **(c)** small-signal equivalent circuit of **(a)**

Example 7.1

We wish to compare the values of g_m , R_{in} , R_o , and A_0 for a CS amplifier that is designed using an NMOS transistor with $L = 0.4\text{ }\mu\text{m}$ and $W = 4\text{ }\mu\text{m}$ and fabricated in a $0.25\text{-}\mu\text{m}$ technology specified to have $\mu_n C_{ox} = 267\text{ }\mu\text{A/V}^2$ and $V'_A = 10\text{ V}/\mu\text{m}$, with those for a CE amplifier designed using a BJT fabricated in a process with $\beta = 100$ and $V_A = 10\text{ V}$. Assume that both devices are operating at a drain (collector) current of $100\text{ }\mu\text{A}$.

Solution

For simplicity, we shall neglect the Early effect in the MOSFET in determining V_{OV} ; thus,

$$I_D = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)V_{OV}^2$$
$$100 = \frac{1}{2} \times 267 \times \left(\frac{4}{0.4}\right)V_{OV}^2$$

resulting in

$$V_{OV} = 0.27\text{ V}$$
$$g_m = \frac{2I_D}{V_{OV}} = \frac{2 \times 0.1}{0.27} = 0.74\text{ mA/V}$$
$$R_{in} = \infty$$
$$r_o = \frac{V'_A L}{I_D} = \frac{10 \times 0.4}{0.1} = 40\text{ k}\Omega$$
$$R_o = r_o = 40\text{ k}\Omega$$
$$A_0 = g_m r_o = 0.74 \times 40 = 29.6\text{ V/V}$$

For the CE amplifier we have

$$g_m = \frac{I_C}{V_T} = \frac{0.1\text{ mA}}{0.025\text{ V}} = 4\text{ mA/V}$$
$$R_{in} = r_\pi = \frac{\beta}{g_m} = \frac{100}{4} = 25\text{ k}\Omega$$
$$r_o = \frac{V_A}{I_C} = \frac{10}{0.1} = 100\text{ k}\Omega$$
$$R_o = r_o = 100\text{ k}\Omega$$
$$A_0 = g_m r_o = 4 \times 100 = 400\text{ V/V}$$

7.2.2. The Intrinsic Gain

The following equations applied to active-loaded CS amplifier:

$$(7.1) R_{in} = \infty$$

$$(7.2) A_{vo} = -g_m r_o$$

Since this is the maximum gain obtainable in a CS or CE amplifier, we refer to it as the **intrinsic gain** and give it the symbol A_o .

$$(7.3) R_o = r_o$$

- There are **three possible expressions for g_m** , two are particularly useful here.

$$(7.10) g_m = \frac{I_D}{V_{OV}/2}$$

$$(7.11) g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L}} \sqrt{I_D}$$

7.2.2. The Intrinsic Gain

The following equations apply to MOSFET:

$$(7.12) \ r_o = \frac{V_A}{I_D} = \frac{V'_A L}{I_D} \quad V'_A : \text{technology-dependent component of the Early voltage}$$

$$(7.13) \ A_0 = \frac{V_A}{V_{OV} / 2}$$

$$(7.14) \ A_0 = \frac{2V'_A L}{V_{OV}}$$

$$(7.14) \ A_0 = \frac{V'_A \sqrt{2(\mu_n C_{ox})WL}}{\sqrt{I_D}}$$

7.2.3. Effect of the Output Resistance of the Current-Source Load

- The **current-source load of the CS amplifier** in Figure 7.1(a) can be implemented using a PMOS transistor biased in the saturation region to provide the required current I , as shown in Figure 7.3(a).

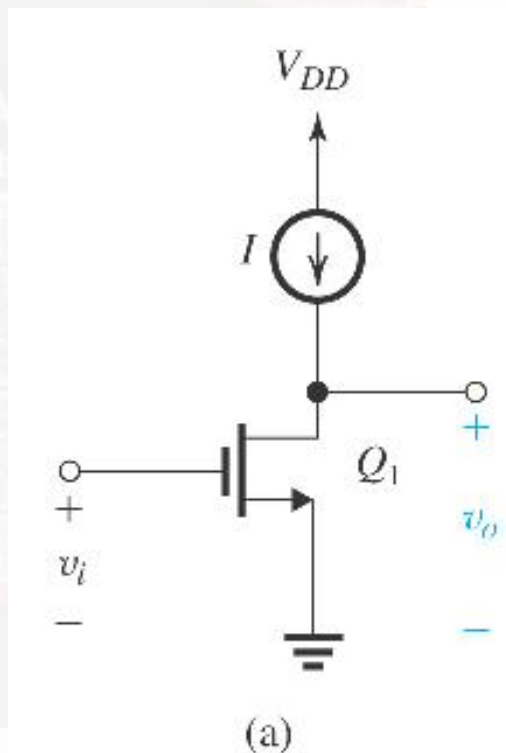


Figure 7.1(a)

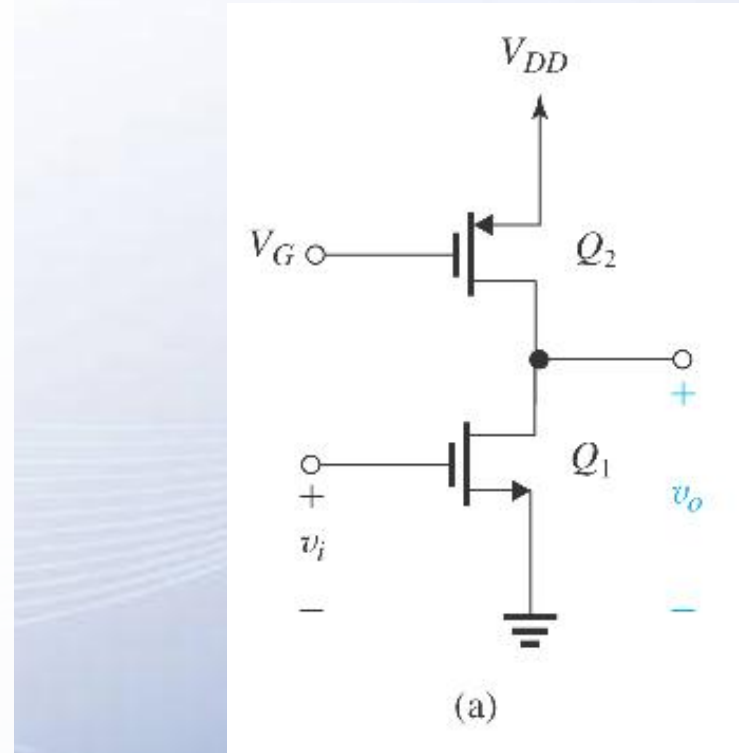
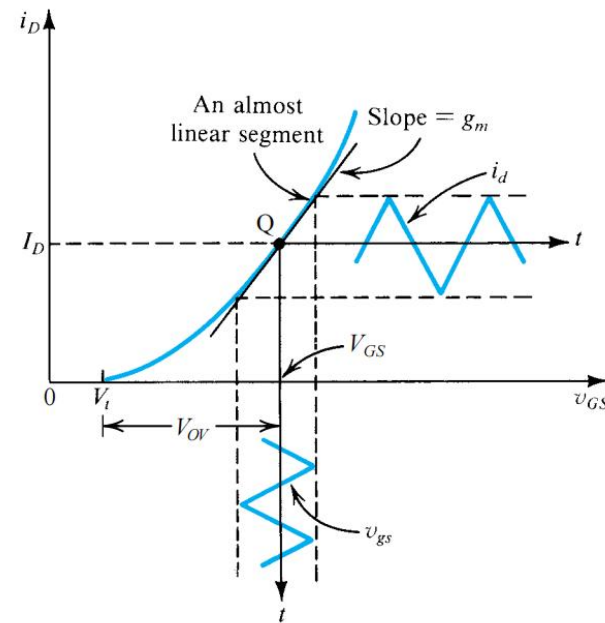


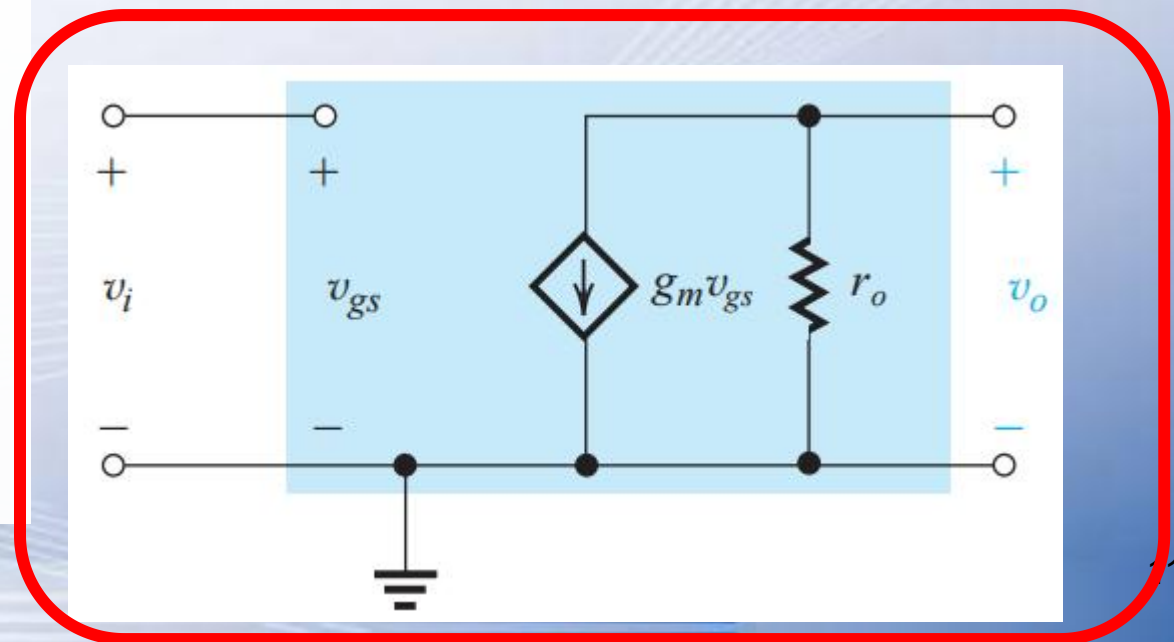
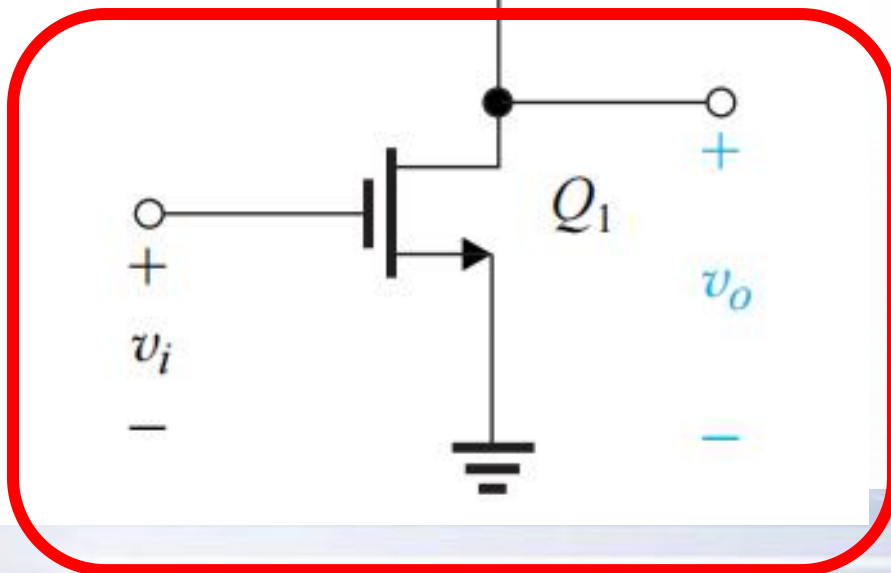
Figure 7.3(a)



$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2 = \frac{1}{2} k_n V_{OV}^2$$

$$g_m \equiv \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS} = V_{GS}}$$

Figure 5.35 Small-signal operation of the MOSFET amplifier.



7.2.3. Effect of the Output Resistance of the Current-Source Load

for PMOS implementation of active-loaded CS amplifier:

large-signal MOSFET model

$$(7.16) \quad I = \frac{1}{2} (\mu_p C_{ox}) \frac{W}{L} [V_{DD} - V_G - |V_{tp}|]^2$$

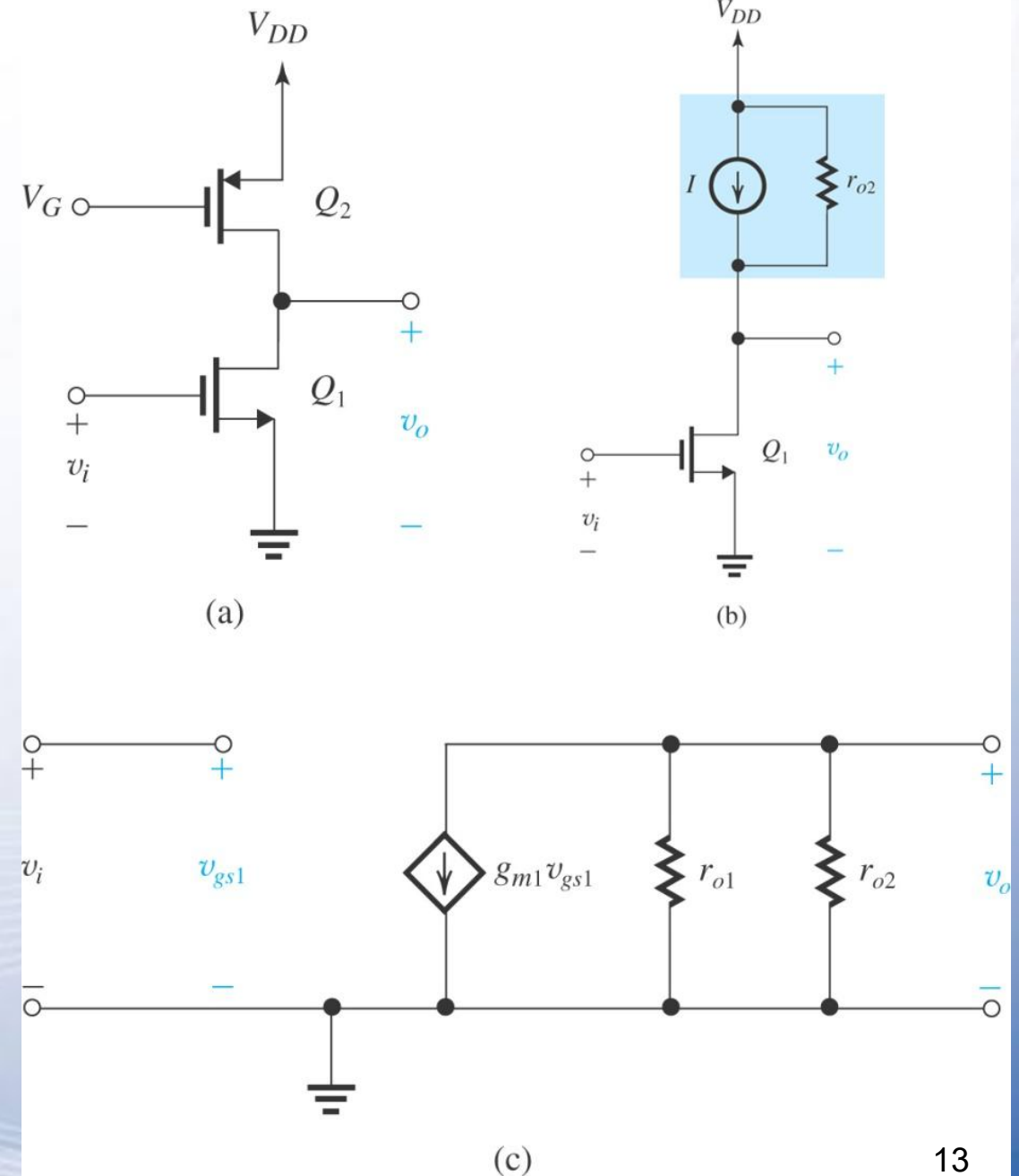
$$(7.17) \quad r_{o2} = \frac{|V_{A2}|}{I}$$

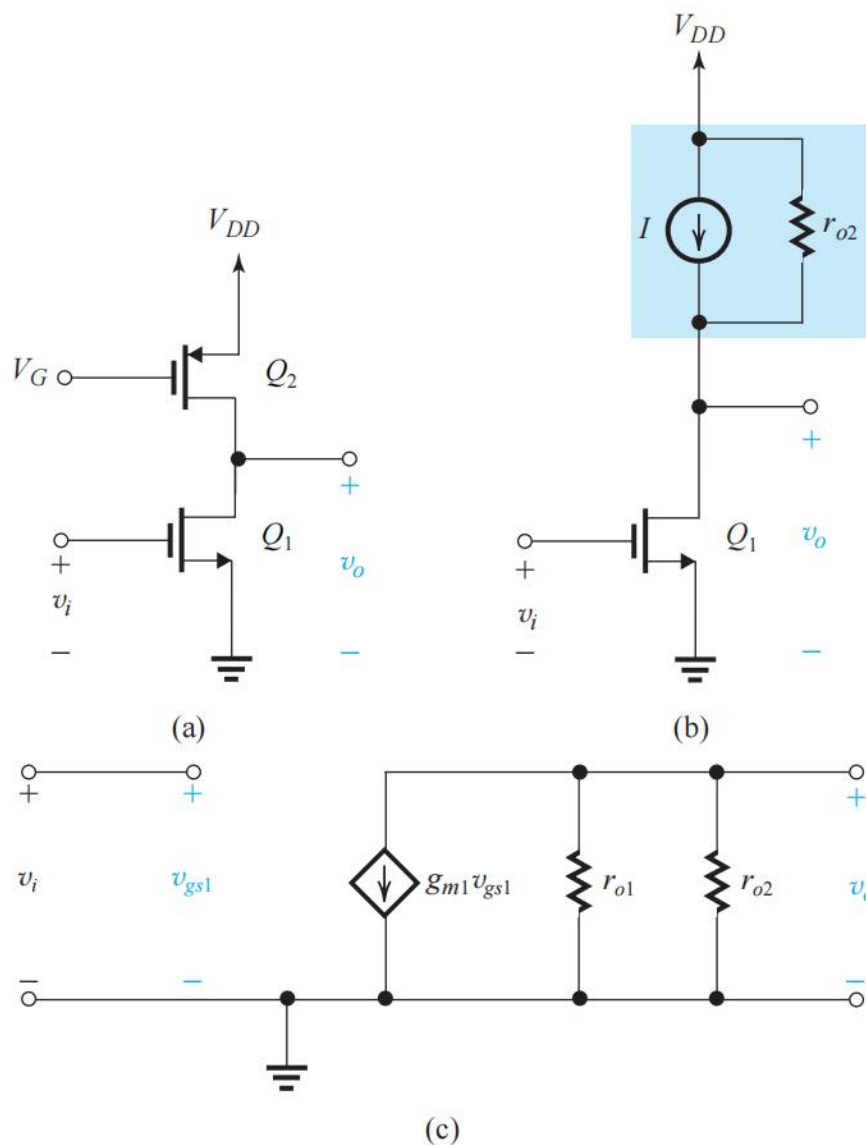
current-source
no longer has
infinite output
resistance

$$(7.18) \quad A_v = \frac{V_o}{V_i} = -g_{m1} (r_{o1} || r_{o2})$$

7.2.3. Effect of the Output Resistance of the Current-Source Load

Figure 7.3 (a): The CS amplifier with the current-source load implemented with a p -channel MOSFET Q_2 ; **(b)** the circuit with Q_2 replaced with its large-signal model; and **(c)** small-signal equivalent circuit of the amplifier.





$$I = \frac{1}{2}(\mu_p C_{ox})\left(\frac{W}{L}\right)_2 [V_{DD} - V_G - |V_{tp}|]^2 \quad (7.16)$$

$$r_{o2} = \frac{|V_{A2}|}{I} \quad (7.17)$$

$$A_v \equiv \frac{v_o}{v_i} = -g_{m1}(r_{o1} \parallel r_{o2}) \quad (7.18)$$

$$r_{o2} = r_{o1}$$

$$A_v = -\frac{1}{2}g_m r_o \quad (7.18')$$

Figure 7.3 (a) The CS amplifier with the current-source load implemented with a p -channel MOSFET Q_2 ; (b) the circuit with Q_2 replaced with its large-signal model; and (c) small-signal equivalent circuit of the amplifier.

Example 7.2

A practical circuit implementation of the common-source amplifier is shown in Fig. 7.4(a). Here the current-source transistor Q_2 is the output transistor of a current mirror formed by Q_2 and Q_3 and fed with a reference current I_{REF} . Current mirrors were briefly introduced in Section 5.7.4 and will be studied more extensively in Sections 7.4 and 7.5. For the time being, assume that Q_2 and Q_3 are matched. Also assume that I_{REF} is a stable, well-predicted current that is generated with a special circuit on the chip. To be able to clearly see the region of v_I over which the circuit operates as an almost-linear amplifier, determine the voltage transfer characteristic (VTC), that is, v_O versus v_I .

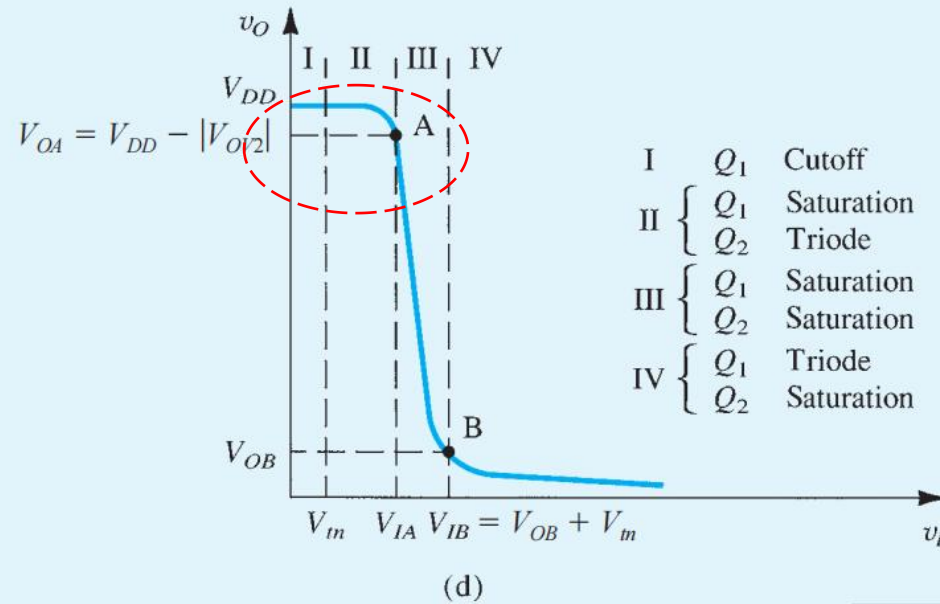
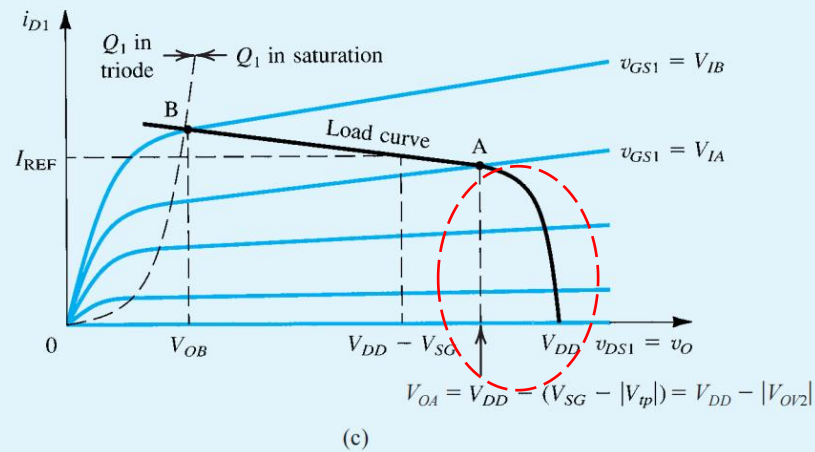
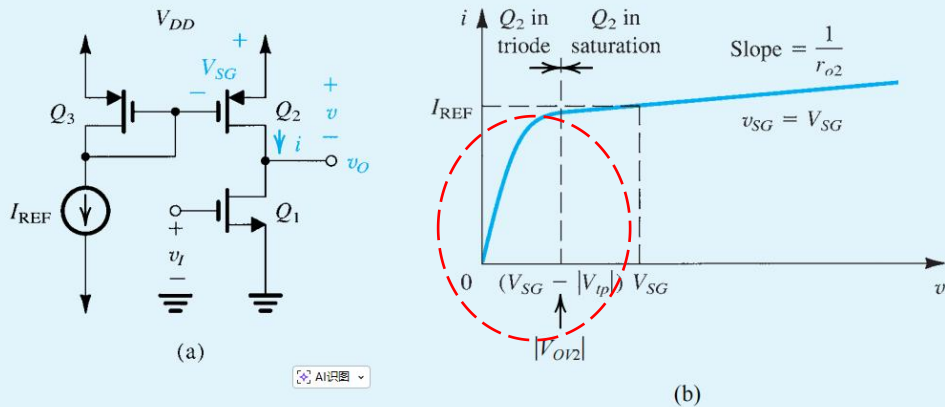


Figure 7.4 Practical implementation of the common-source amplifier: (a) circuit; (b) $i-v$ characteristic of the active-load Q_2 ; (c) graphical construction to determine the transfer characteristic; (d) transfer characteristic.

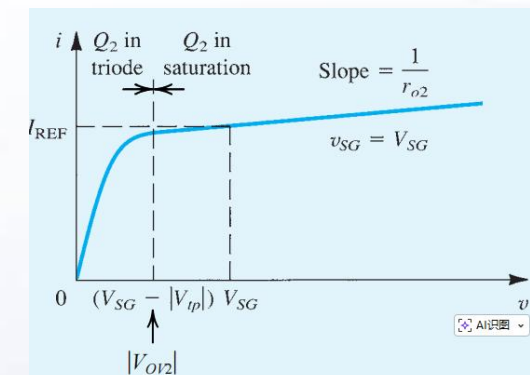
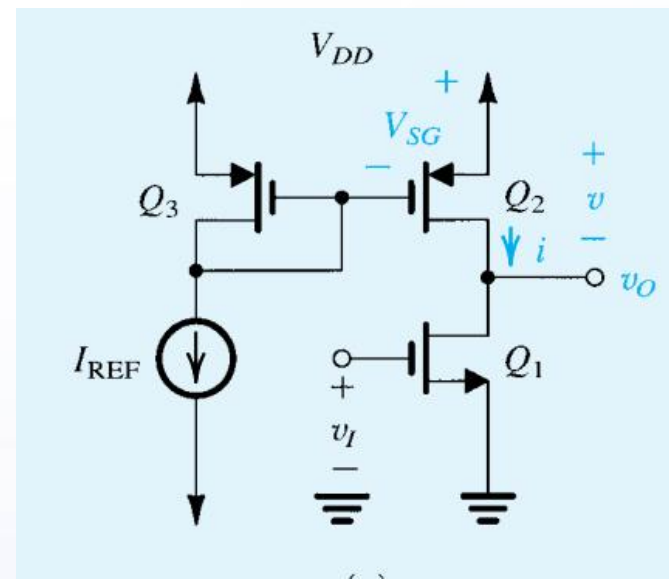
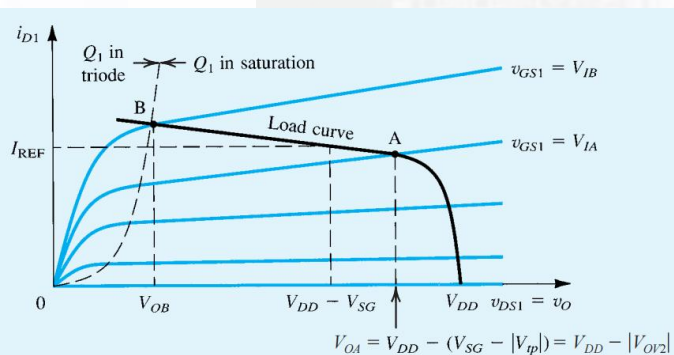
Solution

First we concern ourselves with the current mirror, with the objective of determining the $i-v$ characteristic of the current source Q_2 . Toward that end, we note that the current I_{REF} flows through the diode-connected transistor Q_3 and thus determines V_{SG} of Q_3 , which is in turn applied between the source and the gate of Q_2 . Thus, the $i-v$ characteristic of the current source Q_2 will be the i_D-v_{SD} characteristic curve of Q_2 obtained for $v_{SG} = V_{SG}$. This is shown in Fig. 7.4(b), where we note that i will be equal to I_{REF} at one point only, namely, at $v_{SD2} = V_{SG}$, this being the only point at which the two matched transistors Q_2 and Q_3 have identical operating conditions. We also observe the effect of channel-length modulation in Q_2 (the Early effect), which is modeled by the finite output resistance r_{o2} . Finally, note that Q_2 operates as a current source when v is equal to or greater than $|V_{OV2}| = V_{SG} - |V_{tp}|$. This in turn is obtained when $v_O \leq V_{DD} - |V_{OV2}|$. This is the maximum permitted value of the output voltage v_O .

Now, with the $i-v$ characteristic of the current-source load Q_2 in hand, we can proceed to determine v_O versus v_I . Figure 7.4(c) shows a graphical construction for doing this. It is based on the graphical analysis method employed in Section 5.4.5 except that here the load line is not a straight line but is the $i-v$ characteristic curve of Q_2 shifted along the v_O axis by V_{DD} volts and “flipped around.” The reason for this is that

$$v_O = V_{DD} - v$$

The term V_{DD} necessitates the shift, and the minus sign of v gives rise to the “flipping around” of the load curve.

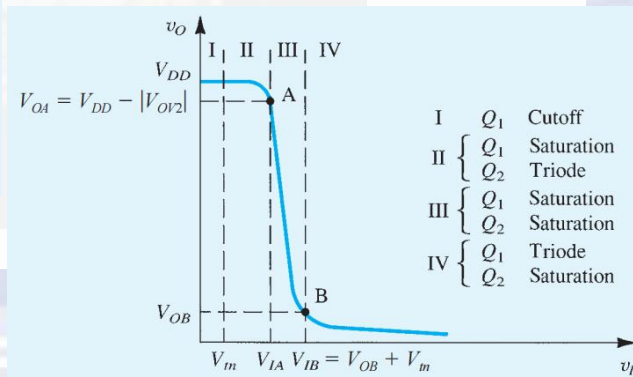


The graphical construction of Fig. 7.4(c) can be used to determine v_O for every value of v_I , point by point: The value of v_I determines the particular characteristic curve of Q_1 on which the operating point lies. The operating point will be at the intersection of this particular graph and the load curve. The horizontal coordinate of the operating point then gives the value of v_O .

Proceeding in the manner just explained, we obtain the VTC shown in Fig. 7.4(d). As indicated, it has four distinct segments, labeled I, II, III, and IV. Each segment is obtained for one of the four combinations of the modes of operation of Q_1 and Q_2 , which are also indicated in the diagram. Note that we have labeled two important break points on the transfer characteristic (A and B) in correspondence with the intersection points (A and B) in Fig. 7.4(c). We urge the reader to carefully study the transfer characteristic and its various details.

Not surprisingly, segment III is the one of interest for amplifier operation. Observe that in region III the transfer curve is almost linear and is very steep, indicating large voltage gain. In region III both the amplifying transistor Q_1 and the load transistor Q_2 are operating in saturation. The end points of region III are A and B: At A, defined by $v_O = V_{DD} - |V_{OV2}|$, Q_2 enters the triode region, and at B, defined by $v_O = v_I - V_{tn}$, Q_1 enters the triode region. When the amplifier is biased at a point in region III, the small-signal voltage gain can be determined as we have done in Fig. 7.3(c). The question remains as to how we are going to guarantee that the dc component of v_I will have such a value that will result in operation in region III. That is why overall negative feedback is needed, as will be demonstrated later.

Before leaving this example it is useful to reiterate that the upper limit of the amplifier region (i.e., point A) is defined by $V_{OA} = V_{DD} - |V_{OV2}|$ and the lower limit (i.e., point B) is defined by $V_{OB} = V_{OV1}$, where V_{OV1} can be approximately determined by assuming that $I_{D1} \approx I_{REF}$. A more precise value for V_{OB} can be obtained by taking into account the Early effect in both Q_1 and Q_2 , as will be demonstrated in the next example.



Example 7.3

Consider the CMOS common-source amplifier in Fig. 7.4(a) for the case $V_{DD} = 3\text{ V}$, $V_{tn} = |V_{tp}| = 0.6\text{ V}$, $\mu_n C_{ox} = 200\text{ }\mu\text{A/V}^2$, and $\mu_p C_{ox} = 65\text{ }\mu\text{A/V}^2$. For all transistors, $L = 0.4\text{ }\mu\text{m}$ and $W = 4\text{ }\mu\text{m}$. Also, $V_{An} = 20\text{ V}$, $|V_{Ap}| = 10\text{ V}$, and $I_{REF} = 100\text{ }\mu\text{A}$. Find the small-signal voltage gain. Also, find the coordinates of the extremities of the amplifier region of the transfer characteristic—that is, points A and B.

Solution

$$g_{m1} = \sqrt{2k'_n \left(\frac{W}{L}\right)_1 I_{REF}}$$

$$= \sqrt{2 \times 200 \times \frac{4}{0.4} \times 100} = 0.63\text{ mA/V}$$

$$r_{o1} = \frac{V_{An}}{I_{D1}} = \frac{20\text{ V}}{0.1\text{ mA}} = 200\text{ k}\Omega$$

$$r_{o2} = \frac{|V_{Ap}|}{I_{D2}} = \frac{10\text{ V}}{0.1\text{ mA}} = 100\text{ k}\Omega$$

Thus,

$$A_v = -g_{m1}(r_{o1} \parallel r_{o2})$$

$$= -0.63(\text{mA/V}) \times (200 \parallel 100)(\text{k}\Omega) = -42\text{ V/V}$$

Approximate values for the extremities of the amplifier region of the transfer characteristic (region III) can be determined as follows: Neglecting the Early effect, all three transistors are carrying equal currents I_{REF} , and thus we can determine the overdrive voltages at which they are operating. Transistors Q_2 and Q_3 will have equal overdrive voltages, $|V_{OV3}|$, determined from

$$I_{D3} = I_{REF} \approx \frac{1}{2}(\mu_p C_{ox}) \left(\frac{W}{L}\right)_3 |V_{OV3}|^2$$

Substituting, $I_{REF} = 100\text{ }\mu\text{A}$, $\mu_p C_{ox} = 65\text{ }\mu\text{A/V}^2$, $(W/L)_3 = 4/0.4 = 10$ results in

$$|V_{OV3}| = 0.55\text{ V}$$

$$|V_{OV3}| = 0.55\text{ V}$$

Thus,

$$V_{OA} = V_{DD} - |V_{OV3}| = 2.45\text{ V}$$

Next we determine $|V_{OV1}|$ from

$$I_{D1} \approx I_{REF} \approx \frac{1}{2}(\mu_n C_{ox}) \left(\frac{W}{L}\right)_1 V_{OV1}^2$$

Substituting, $I_{REF} = 100\text{ }\mu\text{A}$, $\mu_n C_{ox} = 200\text{ }\mu\text{A/V}^2$, $(W/L)_1 = 4/0.4 = 10$ results in

$$V_{OV1} = 0.32\text{ V}$$

Thus, $V_{OB} = V_{OV1} = 0.32\text{ V}$.

More precise values for V_{OA} and V_{OB} can be determined by taking the Early effect in all transistors into account as follows.

First, we determine V_{SG} of Q_2 and Q_3 corresponding to $I_{D3} = I_{REF} = 100\text{ }\mu\text{A}$ using

$$I_{D3} = \frac{1}{2} k'_p \left(\frac{W}{L}\right)_3 (V_{SG} - |V_{tp}|)^2 \left(1 + \frac{V_{SD}}{|V_{Ap}|}\right)$$

Thus,

$$100 = \frac{1}{2} \times 65 \left(\frac{4}{0.4}\right) |V_{OV3}|^2 \left(1 + \frac{0.6 + |V_{OV3}|}{10}\right) \quad (7.19)$$

where $|V_{OV3}|$ is the magnitude of the overdrive voltage at which Q_3 and Q_2 are operating, and we have used the fact that, for Q_3 , $V_{SD} = V_{SG}$. Equation (7.19) can be manipulated to the form

$$0.29 = |V_{OV3}|^2 (1 + 0.09|V_{OV3}|)$$

which by a trial-and-error process yields

$$|V_{OV3}| = 0.53\text{ V}$$

Thus,

$$V_{SG} = 0.6 + 0.53 = 1.13\text{ V}$$

and

$$V_{OA} = V_{DD} - V_{OV3} = 2.47\text{ V}$$

To find the corresponding value of v_I , V_{IA} , we derive an expression for v_O versus v_I in region III. Noting that in region III, Q_1 and Q_2 are in saturation and obviously conduct equal currents, we can write

$$i_{D1} = i_{D2}$$

$$\frac{1}{2} k'_n \left(\frac{W}{L}\right)_1 (v_I - V_{tn})^2 \left(1 + \frac{v_O}{|V_{An}|}\right) = \frac{1}{2} k'_p \left(\frac{W}{L}\right)_2 (V_{SG} - |V_{tp}|)^2 \left(1 + \frac{V_{DD} - v_O}{|V_{Ap}|}\right)$$

Substituting numerical values, we obtain

$$8.55(v_I - 0.6)^2 = \frac{1 - 0.08v_O}{1 + 0.05v_O} \approx (1 - 0.13v_O)$$

which can be manipulated to the form

$$v_O = 7.69 - 65.77(v_I - 0.6)^2 \quad (7.20)$$

This is the equation of segment III of the transfer characteristic. Although it includes v_I^2 , the reader should not be alarmed: Because region III is very narrow, v_I changes very little, and the characteristic is nearly linear. Substituting $v_O = 2.47\text{ V}$ gives the corresponding value of v_I ; that is, $V_{IA} = 0.88\text{ V}$. To determine the coordinates of B, we note that they are related by $V_{OB} = V_{IB} - V_{tn}$. Substituting in Eq. (7.20) and solving gives $V_{IB} = 0.93\text{ V}$ and $V_{OB} = 0.33\text{ V}$. The width of the amplifier region is therefore

$$\Delta v_I = V_{IB} - V_{IA} = 0.05\text{ V}$$

and the corresponding output range is

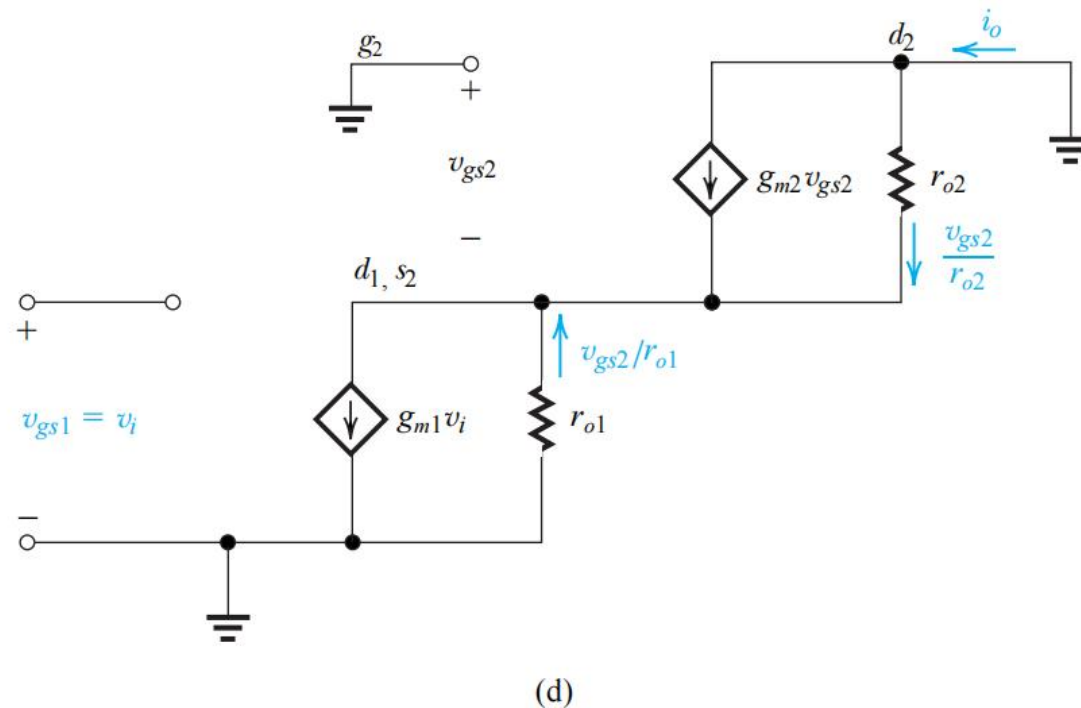
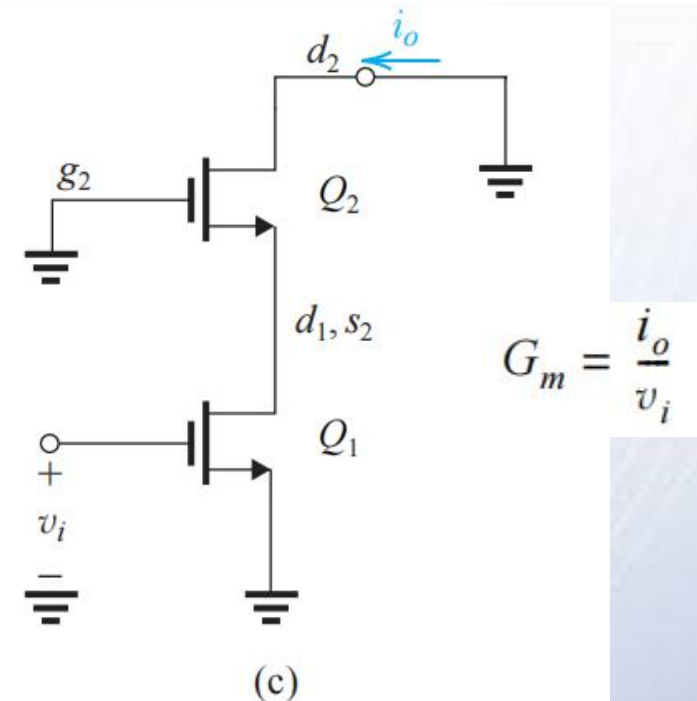
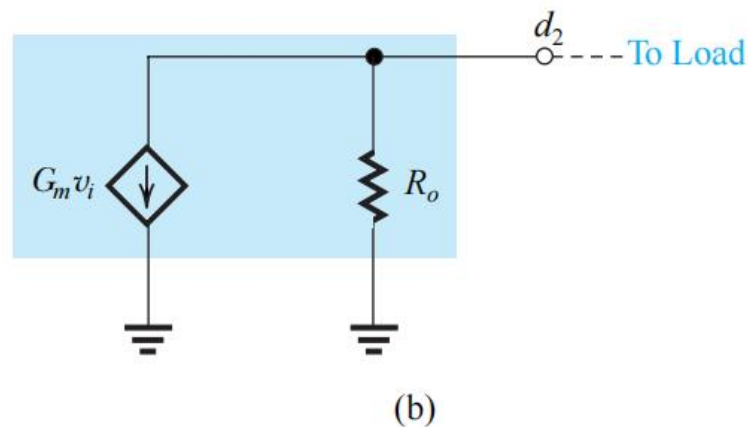
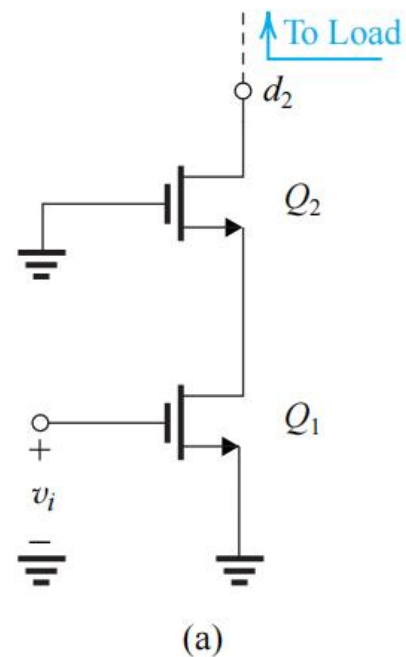
$$\Delta v_O = V_{OB} - V_{OA} = -2.14\text{ V}$$

Thus, the “large-signal” voltage gain is

$$\frac{\Delta v_O}{\Delta v_I} = -\frac{2.14}{0.05} = -42.8\text{ V/V}$$

which is very close to the small-signal value of -42 , indicating that segment III of the transfer characteristic is quite linear.

Figure 7.7 (a) A MOS cascode amplifier prepared for small-signal calculations; (b) output equivalent circuit of the amplifier in (a); (c) the cascode amplifier with the output short-circuited to determine ; (d) equivalent circuit of the situation in (c).



$$g_{m2}v_{gs2} + \frac{v_{gs2}}{r_{o1}} + \frac{v_{gs2}}{r_{o2}} = g_{m1}v_i$$

$$\left(g_{m2} + \frac{1}{r_{o1}} + \frac{1}{r_{o2}}\right)v_{gs2} = g_{m1}v_i$$

Since $g_{m2} \gg (1/r_{o1}), 1/r_{o2}$,

$$g_{m2}v_{gs2} \approx g_{m1}v_i$$

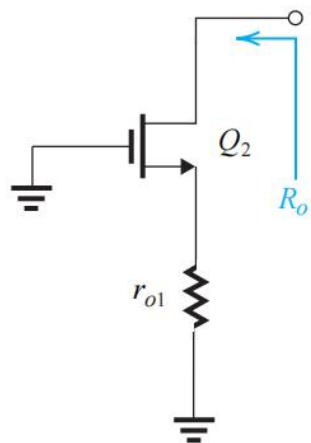
$$i_o = g_{m2}v_{gs2} + \frac{v_{gs2}}{r_{o2}}$$

$$= \left(g_{m2} + \frac{1}{r_{o2}}\right)v_{gs2}$$

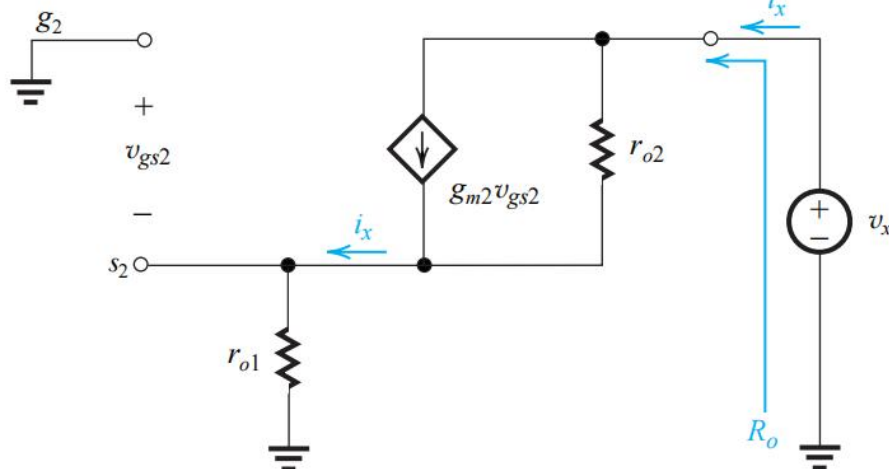
$$i_o \approx g_{m2}v_{gs2}$$

$$i_o = g_{m1}v_i$$

$$G_m = \frac{i_o}{v_i} = g_{m1}$$



(a)



(b)

$$A_{vo} = \frac{v_o}{v_i} = -g_{m1}R_o$$

$$A_{vo} = -(g_{m1}r_{o1})(g_{m2}r_{o2})$$

For the case $g_{m1} = g_{m2} = g_m$ and $r_{o1} = r_{o2} = r_o$,

$$A_{vo} = -(g_m r_o)^2 = -A_0^2$$

$$R_o \equiv \frac{v_x}{i_x}$$

$$-v_{gs2} = i_x r_{o1}$$

$$v_x = (i_x - g_{m2}v_{gs2})r_{o2} + i_x r_{o1}$$

$$v_x = i_x(r_{o1} + r_{o2} + g_{m2}r_{o2}r_{o1})$$

$$R_o = r_{o1} + r_{o2} + g_{m2}r_{o2}r_{o1}$$

$$R_o \approx (g_{m2}r_{o2})r_{o1}$$

$$K = A_{02} = g_{m2}r_{o2}$$

Example 7.4

It is required to design the cascode current-source of Fig. 7.10 to provide a current of 100 μA and an output resistance of 500 $\text{k}\Omega$. Assume the availability of a 0.18- μm CMOS technology for which $V_{DD} = 1.8\text{ V}$, $V_{tp} = -0.5\text{ V}$, $\mu_p C_{ox} = 90\text{ }\mu\text{A/V}^2$ and $V'_A = -5\text{ V}/\mu\text{m}$. Use $|V_{OV}| = 0.3\text{ V}$ and determine L and W/L for each transistor, and the values of the bias voltages V_{G3} and V_{G4} .

Solution

The output resistance R_o is given by

$$R_o = (g_{m3}r_{o3})r_{o4}$$

Assuming Q_3 and Q_4 are identical,

$$\begin{aligned} R_o &= (g_m r_o) r_o \\ &= \frac{|V_A|}{|V_{OV}|/2} \times \frac{|V_A|}{I_D} \end{aligned}$$

Using $|V_{OV}| = 0.3\text{ V}$, we write

$$500\text{ k}\Omega = \frac{|V_A|}{0.15} \times \frac{|V_A|}{0.1\text{ mA}}$$

Thus we require

$$|V_A| = 2.74\text{ V}$$

Now, since $|V_A| = |V'_A| L$ we need to use a channel length of

$$L = \frac{2.74}{5} = 0.55\text{ }\mu\text{m}$$

which is about three times the minimum channel length. With $|V_t| = 0.5\text{ V}$ and $|V_{OV}| = 0.3\text{ V}$,

$$V_{SG4} = 0.5 + 0.3 = 0.8\text{ V}$$

and thus,

$$V_{G4} = 1.8 - 0.8 = 1.0\text{ V}$$

To allow for the largest possible signal swing at the output terminal, we shall use the minimum required voltage across Q_4 , namely, $|V_{OV}|$ or 0.3 V. Thus,

$$V_{D4} = 1.8 - 0.3 = 1.5\text{ V}$$

Since the two transistors are identical and are carrying equal currents,

$$V_{SG3} = V_{SG4} = 0.8\text{ V}$$

Thus,

$$V_{G3} = 1.5 - 0.8 = +0.7\text{ V}$$

We note that the maximum voltage allowed at the output terminal of the current source will be constrained by the need to allow a minimum voltage of $|V_{OV}|$ across Q_3 ; thus,

$$v_{D3\text{max}} = 1.5 - 0.3 = +1.2\text{ V}$$

To determine the required W/L ratios of Q_3 and Q_4 , we use

$$\begin{aligned} I_D &= \frac{1}{2}(\mu_p C_{ox})\left(\frac{W}{L}\right)|V_{OV}|^2\left(1 + \frac{V_{SD}}{|V_A|}\right) \\ 100 &= \frac{1}{2} \times 90 \times \left(\frac{W}{L}\right) \times 0.3^2 \left(1 + \frac{0.3}{2.74}\right) \end{aligned}$$

which yields

$$\frac{W}{L} = 22.3$$

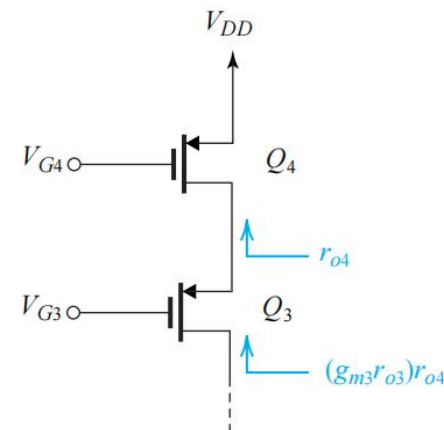


Figure 7.10 Employing a cascode transistor Q_3 to raise the output resistance of the current source Q_4 .



7.4. IC Biasing – Current Sources, Current Mirrors, and Current-Steering Circuits

- Biasing in integrated-circuit design is based on the **use of constant-current sources**.
 - On an IC chip with a number of amplifier stages, a constant dc current (**reference current**) is generated at one location and is then replicated at various other locations for biasing.
 - This is known as **current steering**.
- This approach has the advantage that the **effort expended on generating a predictable and stable reference current** need not be repeated.

7.4.1. The Basic MOSFET Current Source

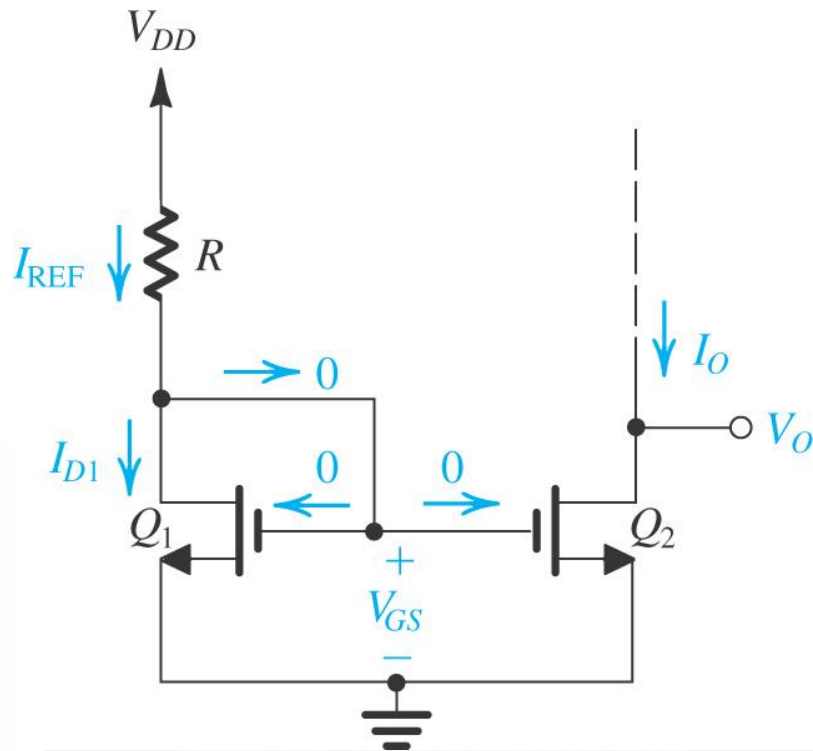


Figure 7.22 Circuit for a basic MOSFET constant-current source. For proper operation, the output terminal, that is, the drain of Q_2 , must be connected to a circuit that ensures that Q_2 operates in saturation.

- Figure 7.22. shows the circuit of a **simple MOS constant-current source**.
- The head of the circuit is transistor Q_1 , the drain of which is shorted to the gate, thereby forcing it to **operate in saturation mode** – equation (7.52).
- The **drain current of Q_1 is supplied by V_{DD}** through resistor R which in most cases is located outside of the IC chip.
- If one considers transistor Q_2 , it is realized that it has **V_{GS} identical to Q_1** – thus (7.54) through (7.59) apply.

7.4.1. The Basic MOSFET Current Source

$$(7.52) I_{D1} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_1 (V_{GS} - V_{tn})^2$$

$$(7.53) I_{D1} = I_{REF} = \frac{V_{DD} - V_{GS}}{R}$$

$$(7.54) I_O = I_{D2} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_2 (V_{GS} - V_{tn})^2$$

$$(7.55) \frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

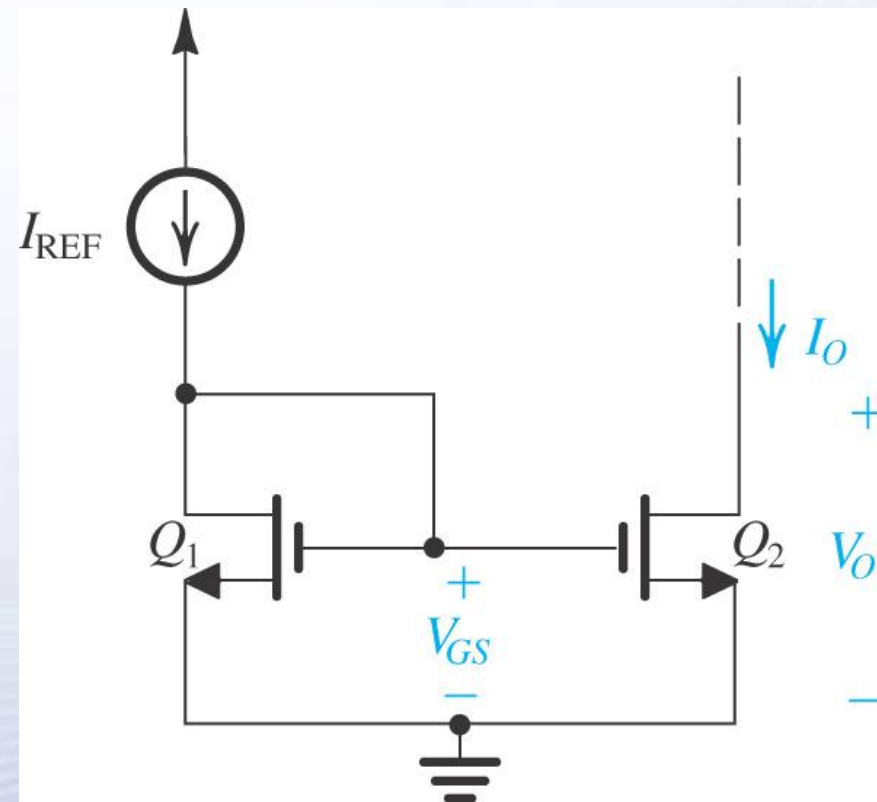


Figure 7.23: Basic MOSFET current mirror.

7.4.1. The Basic MOSFET Current Source

- The combination of Q_1 and Q_2 is referred to as a **current mirror**.
- Figure 7.23 depicts the current-mirror circuit with the input reference current shown as being **supplied by a current source for both simplicity and generality**.
- The **current gain or current transfer ratio** of the mirror is given by (7.55).
- Once a constant current has been generated, **it can be replicated to provide dc bias** or load current for the various stages of the amplifier in an IC.
- **Current mirrors** can be used to achieve this goal.

7.4.2. MOS Current-Steering Circuits

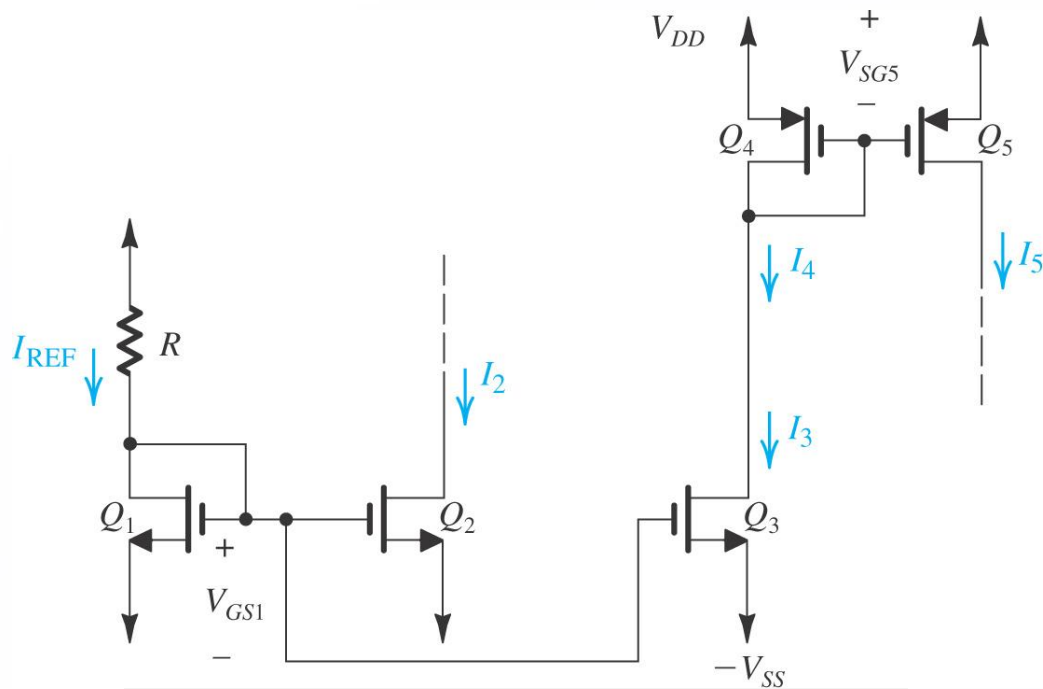


Figure 7.25: A current-steering circuit.

$$(7.60) \quad I_2 = I_{REF} \frac{(W/L)_2}{(W/L)_1}$$

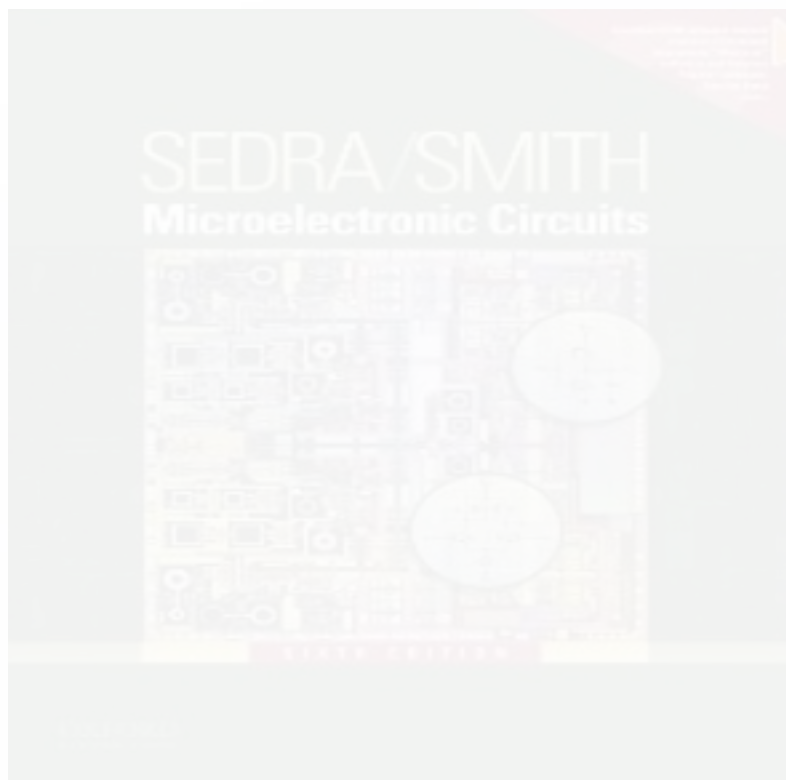
$$(7.61) \quad I_3 = I_{REF} \frac{(W/L)_3}{(W/L)_1}$$

$$(7.62) \quad V_{D2}, V_{D3} \geq -V_{SS} + V_{GS1} - V_{tn}$$

$$(7.63) \quad V_{D2}, V_{D3} \geq -V_{SS} + V_{OV1}$$

$$(7.64) \quad I_5 = I_4 \frac{(W/L)_5}{(W/L)_4}$$

$$(7.65) \quad V_{D5} \leq V_{DD} - |V_{OV5}|$$



Given $V_{DD} = 3\text{ V}$ and using $I_{REF} = 100\text{ }\mu\text{A}$, design the circuit of Fig. 7.22 to obtain an output current whose nominal value is $100\text{ }\mu\text{A}$. Find R if Q_1 and Q_2 are matched and have channel lengths of $1\text{ }\mu\text{m}$, channel widths of $10\text{ }\mu\text{m}$, $V_t = 0.7\text{ V}$, and $k'_n = 200\text{ }\mu\text{A/V}^2$. What is the lowest possible value of V_O ? Assuming that for this process technology, the Early voltage $V'_A = 20\text{ V}/\mu\text{m}$, find the output resistance of the current source. Also, find the change in output current resulting from a $+1\text{-V}$ change in V_O .

Solution

$$I_{D1} = I_{REF} = \frac{1}{2}k'_n \left(\frac{W}{L}\right)_1 V_{OV}^2$$

$$100 = \frac{1}{2} \times 200 \times 10 V_{OV}^2$$

Thus,

$$V_{OV} = 0.316\text{ V}$$

and

$$V_{GS} = V_t + V_{OV} = 0.7 + 0.316 \approx 1\text{ V}$$

$$R = \frac{V_{DD} - V_{GS}}{I_{REF}} = \frac{3 - 1}{0.1\text{ mA}} = 20\text{ k}\Omega$$

$$V_{Omin} = V_{OV} \approx 0.3\text{ V}$$

For the transistors used, $L = 1\text{ }\mu\text{m}$. Thus,

$$V_A = 20 \times 1 = 20\text{ V}$$

$$r_{o2} = \frac{20\text{ V}}{100\text{ }\mu\text{A}} = 0.2\text{ M}\Omega$$

The output current will be $100\text{ }\mu\text{A}$ at $V_O = V_{GS} = 1\text{ V}$. If V_O changes by $+1\text{ V}$, the corresponding change in I_O will be

$$\Delta I_O = \frac{\Delta V_O}{r_{o2}} = \frac{1\text{ V}}{0.2\text{ M}\Omega} = 5\text{ }\mu\text{A}$$

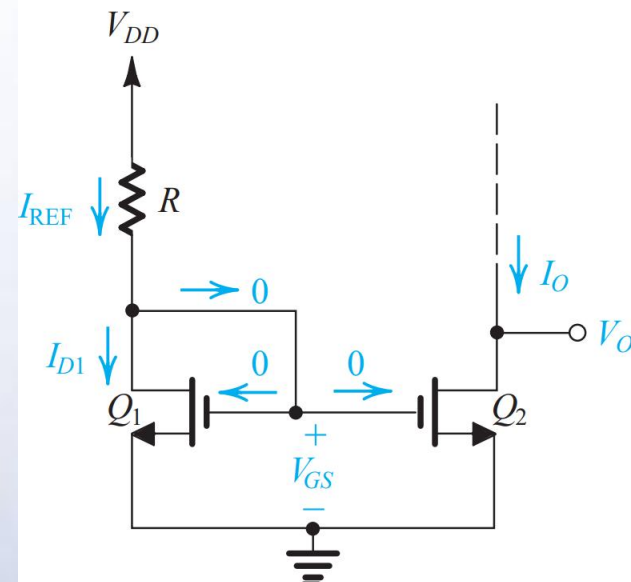


Figure 7.22 Circuit for a basic MOSFET constant-current source. For proper operation, the output terminal, that is, the drain of Q_2 , must be connected to a circuit that ensures that Q_2 operates in saturation.

Summary

- Integrated-circuit fabrication technology offers the circuit designer many exciting opportunities, the most important of which is the large number of inexpensive small-area MOS transistors. An overriding concern for IC designers, however, is the minimization of chip area or “silicon real estate.” As a result, large-valued resistors and capacitors are **virtually absent**.
- The basic gain cell of IC amplifier is the CS (CE) amplifier with a current-source load. For an ideal current-source load (i.e. one with infinite output resistance), the transistor operates in an open-circuit fashion and thus provides the maximum gain possible: $A_{vo} = -g_m r_o = -A_0$.

Summary

- The intrinsic gain A_0 is given by $A_0 = V_A / V_T$ for a BJT and $A_0 = V_A / (V_{OV}/2)$ for a MOSFET. For a MOSFET, A_0 is inversely proportional to $I_D^{1/2}$. See equation 7.15.
- Simple current-source loads reduce the gain realized in the basic gain cell because of their finite resistance (usually comparable to the value of r_o of the amplifying transistor).
- To raise the output resistance of the CS transistor, we stack a CG on top. This is cascoding. The CG or CB transistor in the cascode passes the current $g_{m1}v_i$ provided by the CS or CE transistor.

Summary

- A MOS cascode amplifier operating with an ideal current source load achieves a gain of $(g_m r_o)^2 = A_0^2$.
- Biasing in integrated circuits utilizes current sources. As well, current sources are used as load devices. Typically an accurate and stable reference current is generated and then replicated to provide bias current for the various amplifier stages on the chip. The heart of the **current-steering** circuitry utilized to perform this function is the current mirror.
- The MOS current mirror has a current transfer ratio of $(W/L)_2/(W/L)_1$.
- MOS mirrors of the basic type has a finite output resistance equal to r_o of the output device. Also, for proper operation, a voltage of at least 0.3V is required across the output transistor of a simple bipolar mirror ($|V_{OV}|$ for the MOS case).

7.12 Figure P7.12 shows an IC MOS amplifier formed by cascading two common-source stages. Assuming that $V_{An} = |V_{Ap}|$ and that the biasing current sources have output resistances equal to those of Q_1 and Q_2 , find an expression for the overall voltage gain in terms of g_m and r_o of Q_1 and Q_2 .

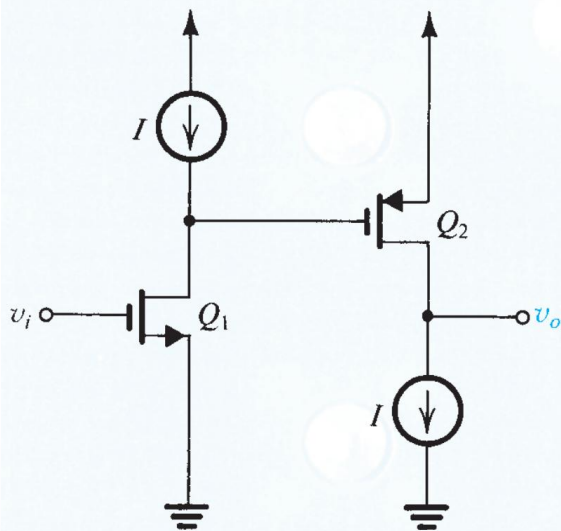


Figure P7.12

****7.18** The MOSFETs in the circuit of Fig. P7.18 are matched, having $k'_n(W/L)_1 = k'_p(W/L)_2 = 1 \text{ mA/V}^2$ and $|V_t| = 0.5 \text{ V}$. The resistance $R = 1 \text{ M}\Omega$.

- For G and D open, what are the drain currents I_{D1} and I_{D2} ?
- For $r_o = \infty$, what is the voltage gain of the amplifier from G to D? [Hint: Replace the transistors with their small-signal models.]
- For finite r_o ($|V_A| = 20 \text{ V}$), what is the voltage gain from G to D and the input resistance at G?
- If G is driven (through a large coupling capacitor) from a source v_{sig} having a resistance of $100 \text{ k}\Omega$, find the voltage gain v_d/v_{sig} .
- For what range of output signals do Q_1 and Q_2 remain in the saturation region?

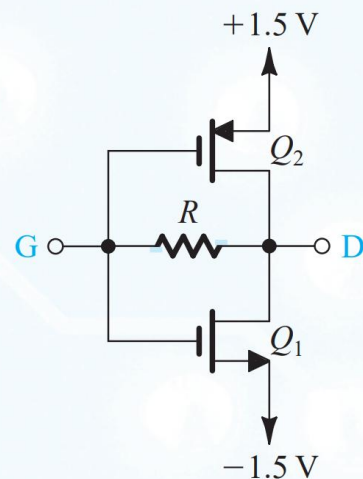


Figure P7.18

D 7.51 The current-steering circuit of Fig. P7.51 is fabricated in a CMOS technology for which $\mu_n C_{ox} = 200 \text{ }\mu\text{A/V}^2$, $\mu_p C_{ox} = 80 \text{ }\mu\text{A/V}^2$, $V_{tn} = 0.6 \text{ V}$, $V_{tp} = -0.6 \text{ V}$, $V'_{An} = 10 \text{ V/}\mu\text{m}$, and $|V'_{Ap}| = 12 \text{ V/}\mu\text{m}$. If all devices have $L = 0.8 \text{ }\mu\text{m}$, design the circuit so that $I_{\text{REF}} = 20 \text{ }\mu\text{A}$, $I_2 = 100 \text{ }\mu\text{A}$, $I_3 = I_4 = 20 \text{ }\mu\text{A}$, and $I_5 = 50 \text{ }\mu\text{A}$. Use the minimum possible device widths needed to achieve proper operation of the current source Q_2 for voltages at its drain as high as $+1.3 \text{ V}$ and proper operation of the current sink Q_5 with voltages at its drain as low as -1.3 V . Specify the widths of all devices and the value of R . Find the output resistance of the current source Q_2 and the output resistance of the current sink Q_5 .

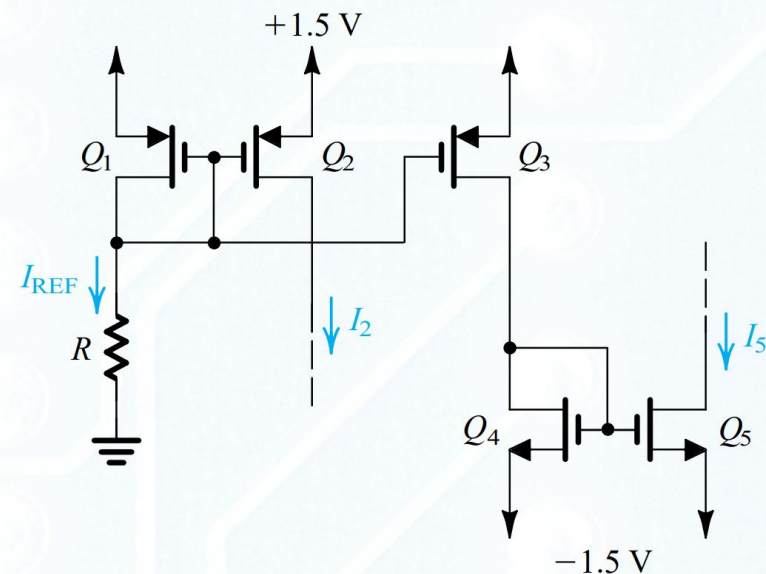


Figure P7.51